

Phytoplankton uptake by a resilient urban estuary bivalve, *Mytilopsis leucophaeata*

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Abstract

Urban estuaries face numerous problems stemming from excess nutrients from stormwater runoff. Baltimore Harbor, an urban estuary, has intense and frequent phytoplankton blooms that can lead to hypoxic and anoxic zones, and ultimately fish kills. Efforts to reduce the nutrients, that fuel algae blooms, focus on point sources and not remediation once the nutrients have entered the water. In water removal of nutrients is often in the form of phytoplankton uptake by bivalve suspension feeders. Commercial bivalve species have been studied for their ability to remove phytoplankton and sequester nutrients in the coastal zone. In urban estuaries, commercially important bivalves often do not naturally occur, or urban pollution would make them unsuitable for human consumption. Therefore, we set out to determine if a locally abundant non-commercial mussel, *Mytilopsis leucophaeata*, could reduce phytoplankton levels and how environmentally relevant temperature and salinity may impact clearance rates. Our study developed a standard method to quantify phytoplankton removal of two cultured phytoplankton species (*Isochrysis sp.* and *Chaetoceros sp.*), by measuring both in vivo and extracted chlorophyll. We then evaluated the ability of the mussels to remove algae under varied temperatures and salinity that are typical for Baltimore Harbor. Our results show that at lower temperatures there is a lower clearance rate, and salinity did not significantly affect the clearance rates of the mussels. Overall, our results show potential for the use of this native non-commercial bivalve to reduce phytoplankton blooms in Baltimore Harbor.

Introduction

Estuaries are the interface between land and sea and are foci of human activities. About 40% of the US population lives in coastal counties with the population density in shoreline counties four times higher than the national average (Crossett et al. 2013). Runoff from coastal development causes intense stress on urban estuaries, which are already heavily modified by historical and ongoing land uses. The understanding of estuary ecology was initially developed in un-impacted areas, and while the physical impacts of urbanization are also well-studied, their effects on the ecology of urban estuaries are not fully understood (Graells et al. 2021; Pickett et al. 2017). All urban estuaries share the stress of intense nutrient inputs, leading to a cycle of phytoplankton blooms, also called algae blooms, that can cause anoxic and hypoxic zones in the water and shade submerged vegetation (Bricker et al. 2008; Kemp et al. 2005; Lefcheck et al. 2017). Baltimore Harbor is one of many urban estuaries struggling with increased nutrients from stormwater runoff that causes algae blooms. While there are ongoing efforts to reduce nutrient inputs from point and non-point sources (Maryland Department of the Environment 2022), there is an opportunity to develop in-water practices to reduce algae levels in Baltimore Harbor. One under-utilized strategy is to enhance the inherent ability of the urban estuarine ecosystem to remove excess algae through the recruitment of native bivalve species.

From habitat creation to water quality improvements, bivalves provide numerous ecosystem services. Research has focused primarily on commercially valuable bivalve species such as oysters and mussels (Cubillo et al. 2023; Van Den Burg et al. 2022; Kreeger et al. 2018). In the Chesapeake Bay, where oyster

aquaculture is an economically important industry, researchers and managers have quantified ecosystem services of oysters to create a nutrient trading system (Cornwell et al. 2016; Kellogg et al. 2013). However, oysters may not grow well in polluted, low salinity, and often hypoxic urban environments like Baltimore Harbor, and there is a public health risk should people harvest and consume oysters grown for ecosystem services and restoration purposes. Therefore, using an alternative bivalve species in these urban estuaries to reduce phytoplankton and nitrogen levels would reduce potential harm to humans.

One promising bivalve for this type of ecosystem service in Baltimore Harbor is the Dark False Mussel, *Mytilopsis leucophaeata*. *M. leucophaeata* is a dreissenid mussel native to the Gulf of Mexico that has spread to estuaries on the east coast of North America and the lagoons in Mexico (Kennedy 2011b). The mussels are small, about 1–2 cm long, and will settle on any hard substrate in the water. They can tolerate salinities from 5 to 20 and a wide temperature range (Kennedy 2011a; Rajagopal et al. 2005). In the Chesapeake Bay, when salinity has been low, irruptions of *M. leucophaeata* are often seen in urbanized areas (Goldman 2007). *M. leucophaeata* has expanded past its native range and become an invasive pest in South America and the Netherlands. Research in Brazil, where *M. leucophaeata* is invasive, has shown that the mussel holds promise for improving water quality (Neves et al. 2020; Rodrigues et al. 2023). For the past decade this species has been abundant in Baltimore Harbor and so is of interest as a nature-based mitigation for algae blooms and nutrient extraction.

Baltimore Harbor is a complex and dynamic body of water. Tidal forces and flash freshwater inputs cause shifts in salinity (2–15 ppt), and temperature (summer peaks as high as 30°C). Historic dredging for shipping has changed the landscape away from a natural soft shoreline with submerged aquatic vegetation to a hardened shoreline devoid of underwater structured habitat or shallows. Despite this, many species still inhabit Baltimore Harbor, including noticeable populations of iconic Chesapeake Bay life, such as striped bass, menhaden, and blue crabs, and the occasional cow-nosed rays (www.inaturalist.org/places/baltimore-city-county#/). However, like many urban estuaries, Baltimore Harbor faces excessive stormwater runoff that leads to eutrophication. Eutrophication fuels algae blooms, which can exceed 300 $\mu\text{g L}^{-1}$ (<https://eyesonthebay.dnr.maryland.gov/contmon/aquarium.cfm>). These algae blooms can lead to anoxic and hypoxic zones that make the water unsuitable for the fish and crabs to live there. Baltimore Harbor has a recurring population of *M. leucophaeata* making these mussels an ideal species to study for ecosystem services, beginning with their ability to reduce algae levels in the water.

This study aimed to measure the ability of *M. leucophaeata* to reduce algae levels and how environmental factors impact clearance rates. We conducted three experiments to better understand the filtering capabilities of *M. leucophaeata*. First, we conducted a baseline phytoplankton removal experiment with two different cultured algae species to create a standard methodology. Second, we measured clearance rates at three different temperatures that reflect conditions experienced in the Baltimore Harbor. Third, we investigated clearance rates at three different salinities that fall in the range found in Baltimore Harbor. We hypothesize that the mussels should be able to reduce phytoplankton in

all situations. These results provide quantitative information needed to evaluate how *M. leucophaeata* can contribute to phytoplankton and thus nutrient removal from Baltimore Harbor.

Methods

We performed three different experiments to test the clearance rate of *M. leucophaeata*: 1) mussels and different cultured phytoplankton species, 2) mussels at three different temperatures, and 3) mussels at three different salinities. The first experiment tested two algae species and was conducted at 22°C in 10 ppt artificial seawater (ASW). The second experiment examined the effect of temperatures (10°C, 22°C, and 30°C) on the mussel clearance rate. The third experiment examined the effect of salinities (5 ppt, 10 ppt, and 15 ppt) at 22°C on mussel clearance rate. Experimental temperatures and salinities were all chosen based on the range of these parameters recorded by the Eyes on the Bay continuous water quality sondes in Baltimore Harbor from 2018 to 2022 (**Fig. S1, Fig. S2**).

Animal Care and Collection:

Naturally recruited *M. leucophaeata* were collected from rigid substrates in 0.25 to 1.0 m depth at three locations in Baltimore Harbor: 1) The Downtown Sailing Center, 2) the National Aquarium, and 3) the Institute of Marine and Environmental Technology (IMET) (**Table S1**). Mussels were brought back to the lab to acclimate to experimental conditions for a minimum of 10 days before starting any experiment. In this time, mussels were detached from their original substrate and reattached to plastic surfaces, taking advantage of their ability to form new byssal threads. For the algae species experiment, mussels (1.4 +/- 0.08 cm length and about 0.08 +/- 0.01 g of dry tissue weight) were allowed to attach to 10 cm diameter acrylic discs. For the salinity and temperature experiment, mussels (1.5 +/- 0.01 cm length, and 0.06 +/- 0.003 g dry tissue weight) were attached to ~ 6 cm square polypropylene panels. Approximately 15 +/- 3 mussels were used in the algae species experiment and 10 +/- 2 mussels in each replicate for the temperature and salinity experiments. During acclimatization, water was changed every other day with Baltimore Harbor water equilibrated to ambient laboratory temperature.

Salinity and temperature experiments used water from Baltimore Harbor, which was adjusted to target temperature and salinity using water baths and ASW produced by the Aquaculture Research Center (ARC) at IMET. For the temperature and salinity experiments, mussels were acclimated to the appropriate temperatures and salinity for 10 days before the experiment. To adjust the temperature, water baths were gradually shifted to the target temperatures of 10°C and 30°C in 2-degree increments every other day. Approximately 22°C conditions were maintained at room temperature without a water bath. For the salinity experiment, we acclimated mussels to 5 ppt, 10 ppt, and 15 ppt using ASW. Using a similar method as for the temperature experiment, we gradually added the ASW of the corresponding salinity to harbor water to acclimate the mussels. ASW was made by diluting 35 ppt ASW with deionized water from ARC.

Experimental Design of Phytoplankton Uptake Experiments:

Phytoplankton used in the experiments were obtained from the University of Maryland Center for Environmental Science (UMCES) Horn Point Laboratory Oyster Hatchery in Cambridge, MD. Two algae species were compared, the flagellate *Isochrysis* sp. and the diatom *Chaetoceros* sp. For temperature and salinity experiments, only *Isochrysis* sp. was used. *Isochrysis* is about 4–8 microns long and 4–6 microns wide, and *Chaetoceros* is about 4–9 microns long and 4–10 microns wide. Algae cultures were maintained for up to two weeks at room temperature (~ 22°C) in f/2 medium at 10 ppt salinity under constant illumination from a 60-watt equivalent white LED light at a distance of 30–50 cm. To calculate the algae needed for the experiment, a Lugol's fixed suspension was counted on a hemocytometer, and a standard curve of cell counts versus in vivo chlorophyll (IVCH) was constructed, and then used to calculate the amount of algae culture needed for a target of 4000 relative fluorescence units (RFU). Due to the variable density and age of the algae cultures, the target of 4000 RFUs was not always met; we detail any deviances below.

Before the start of a clearance rate experiment, mussels were placed in fresh ASW for a minimum of 12 hours. All experiments had four replicates per treatment conditions: 1) algae and mussels (Mussels), 2) algae and no mussel (No Mussels), and 3) mussel-only control. All experiments were conducted in 2-liter (phytoplankton species experiment) or 1-liter (temperature and salinity experiment) polypropylene containers. Water was collected at five time points for the algae species experiment and four time points for the temperature and salinity experiment. Specifications of those time points are listed below.

Chlorophyll measurements:

In vivo chlorophyll (IVCH) was measured using a Turner Design Aquaflour® and measurements were recorded in relative fluorescence units (RFU). In addition to the IVCH measurements, for the algae species experiment we collected water at the T1 (0 minutes), T4 (360 minutes), and T5 (1350 minutes) periods for measurements of extracted chlorophyll. The amount collected was either 4, 10, or 15 ml, and amount filtered was increased using ASW so that the total volume filtered was 15 ml. For extracted chlorophyll measurements, samples (adjusted 15 ml) were filtered onto GF/C filters (Whatman) and stored at -20°C until processing. We sent the frozen samples to the Chesapeake Biological Laboratory Nutrient Analytical Services Laboratory (Solomons, MD) for chlorophyll-a quantification using the EPA 445.0, SM10200H.3 method.

Phytoplankton Species Experiment:

Both *Isochrysis* sp. and *Chaetoceros* sp. phytoplankton cultures were used in this experiment. Mussels were collected from the Aquarium East site (**Table S1**) in April 2023. At the beginning of the experiment, the mussel discs were suspended in individual containers with 1000 ml ASW to acclimate for an hour. Phytoplankton culture was added to a target of 4000 RFUs based on the hemocytometer cell counts and standard curve measurements. Approximately ten minutes after adding the algae (timepoint T1), we started taking water samples for IVCH measurements and chlorophyll extractions. We collected additional water samples at 2, 4, 6, and 22.5 hours (timepoints T2 – T5). IVCH and Lugol's storage

occurred at all the water collections. Water samples for later chlorophyll extractions were collected at the T1 (10 minutes), T4 (6 hours), and T5 (22.5 hours) time points.

Temperature Experiment:

Temperature records in Baltimore Harbor were examined from 2018 to 2022 using the high-frequency Eyes on the Bay dataset from the Aquarium East location (<https://eyesonthebay.dnr.maryland.gov/contmon/aquarium.cfm>). We plotted a histogram of the continuous temperature data rounded to the nearest whole number to visualize the temperature range that the mussels experience. The data show that there is a mean of 16°C and a range from 0°C to 31°C (Fig. S1). We selected temperatures of 10°C, 22°C, and 30°C as they fell within the range that naturally occurs. We conducted two trials to test the effect of temperature on clearance rates of *M. leucophaeata*. Both trials occurred in July 2023 using mussels also collected in July 2023. The first trial used 800 ml of water and the second trial used 500 ml, to accommodate for the differences in the density of the *Isochrysis* sp. cultures.

At the beginning of each experiment, mussels affixed to plastic panels were placed in the water to acclimate for 1 hour. After that, the T0 time point was collected for IVCH measurements. Then *Isochrysis* was added to the containers, and water was collected for IVCH measurements at 10 minutes, 2 hours, 4 hours, and 6 hours (T1-T4 respectively).

Salinity Experiment:

Experimental salinities were chosen in the same way as in the temperature experiment described above. We used the Eyes on the Bay data to determine that the 5-year mean salinity was 7 ppt and the total salinity ranged from 1 to 17 ppt (Fig. S2). Salinities of 5, 10, and 15 ppt were picked for the experiments. Like with the temperature experiment, two separate trials were performed. Trial 1 was conducted in July 2023 and used mussels collected in July 2023. Trial 2 occurred in October 2023 with mussels collected at the end of September 2023. The first trial used 500 ml of water and the second trial used 800 ml. This was to normalize the density of algae stocks used in the experiment. Experimental conditions and procedures for collecting water samples and experiment setup were the same as the temperature experiment.

Clearance rate calculation:

Clearance rates (R_c) for the salinity and temperature experiments were calculated using the following equation (Jacobs et al. 2015):

$$R_c = \frac{V}{nt} \left(\ln \left(\frac{C_0}{C_t} \right) - \ln \left(\frac{C_{0'}}{C_{t'}} \right) \right)$$

Where V = volume in liters, n = number of mussels, t = time in hours, C_0 = concentration at the beginning for the treatment, C_t = concentration at the end for the treatment, $C_{0'}$ = concentration at the beginning for

the control, and C_t = concentration at the end for the control. We calculated the clearance rates using the first timepoint (T1, 10 minutes) and the final time point (T4, 6 hours).

Statistics:

All statistical tests were run in R version 4.2.1 and used packages “Stats”, “Rstatix”, and “car.” Algae species experiment data were analyzed by algae species type for all collected data. IVCH data for the temperature experiment was analyzed as a group. Since salinity experiments were conducted at separate times of the year, data were analyzed by trial.

To assess the effect of the Mussel or No Mussels treatment on IVCH levels we performed a Kruskal-Wallis test since data was not parametric. To test for the effect of time on IVCH levels, we performed a one-way repeated measures ANOVA and a post-hoc pairwise t-test to test for individual comparisons. Each treatment group, Mussels or No Mussels, was tested independently. If assumptions were violated, we instead used the Friedman Rank Sum test.

To test if treatment (Mussels or No Mussels) affected total extracted chlorophyll concentrations, we performed a Kruskal-Wallis test. To test for the effect of time on extracted chlorophyll concentrations, we performed a repeated measures ANOVA and a post-hoc pairwise t-test. If data was non-parametric, a Friedman test was performed instead. For the control *Isochrysis* group, there was a missing data point with the first time point, so we removed the corresponding data point from the two remaining time points for the analysis.

To test for the effect of salinity or temperature on the IVCH-based clearance rate, we used a one-way ANOVA followed by a Tukey’s HSD test. We used a Type III ANOVA for the salinity Trial 1 data due to a missing data point with the 5 ppt treatment. All other ANOVAs were Type II.

Results

Phytoplankton Species Experiment:

The presence of mussels had a significant effect on the average in vivo and extracted chlorophyll levels for suspensions of both *Isochrysis* and *Chaetoceros*. Mussels had a significant effect on IVCH values at the end of the time series for both *Isochrysis* and *Chaetoceros* (Fig. 1. Kruskal-Wallis Test *Isochrysis* – $p = 4.42\text{E-}05$, $n = 4$; *Chaetoceros* – $p = 3.487\text{E-}05$, $n = 4$). The IVCH values changed significantly over time in both Mussel and No Mussel treatment groups for *Isochrysis* (repeated measures ANOVA Mussels – $p = 7.32\text{E-}11$, $n = 4$; No Mussels – $p = 0.000806$, $n = 4$). Pairwise comparisons show that in the Mussel treatment, there was a significant difference between the first time point (T1) and the other four time points (Pairwise t-test T2, $p = 0.006$, $n = 4$; T3, $p = 0.003$, $n = 4$; T4, $p = 0.004$, $n = 4$; and T5, $p = 0.004$, $n = 4$), but no other pairwise comparison was significant. A similar pattern is observed with the No Mussel treatment where the first timepoint (T1) is statistically different than the T2, T3, and T4 time points

(Pairwise t-test T2, $p = 0.022$, $n = 4$; T3, $p = 0.03$, $n = 4$; T4, $p = 0.014$, $n = 4$). Furthermore, the T2 and T4 time points were statistically different from each other for the No Mussels treatment ($p = 0.03$, $n = 4$).

Looking at the *Chaetoceros* results, time had a significant effect on IVCH levels for the Mussel treatment (Kruskal-Wallis Test Mussels – $p = 1.8E-05$, $n = 4$), and in the No Mussel treatment timepoint was not significant. Pairwise comparisons of the mussel treatment show significant differences between the T1 and the T3 (4 hours) and T5 (22.5 hours) timepoint (T3, $p = 0.026$; T5, $p = 1.76E-04$).

Mussels had a significant effect on the extracted chlorophyll levels for both algae species (Fig. 2.; Kruskal-Wallis Test *Isochrysis* – $p = 0.012$; *Chaetoceros* – $p = 0.002$). Overall, time had a significant effect on extracted chlorophyll levels when mussels were present (Friedman's Rank Sum Test *Isochrysis* – $p = 0.039$, $n = 3$; Repeated Measures ANOVA *Chaetoceros* – $p = 0.001$, $n = 4$). However, pairwise comparisons between timepoints in the *Isochrysis* treatment did not show any significant differences; note however the loss of a replicate meant that comparisons were made for triplicate samples instead of quadruplicates. Comparisons between time points for the *Isochrysis* No Mussel treatment were also not significant. For *Chaetoceros*, the Mussel treatment showed significant differences between the T1 and T4 (6 hours) time points ($p = 0.043$, $n = 4$), and the No Mussel treatment showed no significant differences.

Temperature experiments:

Mussels were able to reduce IVCH at all temperatures tested (**Fig. S3**; Kruskal-Wallis Trial 1 – $p = 1.99E-13$, $n = 12$; Trial 2 – $p = 8.68E-14$, $n = 12$). For Trial 2, the mussel treatment data was log-transformed to meet normality assumptions. IVCH levels changed significantly over time for the mussel treatment in both Trial 1 and 2 but did not change significantly in the No Mussels treatments (Friedman's Rank Sum Test Trial 1 – $p = 6.94E-16$, $n = 12$; Repeated Measured ANOVA Trial 2 – $p = 1.54E-05$, $n = 12$). Indeed, all pairwise comparisons of time points for the Mussel treatments proved to be significant (**Table S2**), and no significant pairwise comparison was found for the No Mussel treatments. These results parallel the initial findings from the algae species experiment that mussels alter IVCH levels of phytoplankton cultures.

To analyze the effect of temperature on IVCH-based clearance rate, a one-way ANOVA was performed. Since both trials took place during the summer, the data were pooled together. Initially starting algae concentration (measured as IVCH) was included in the model due to the differing starting concentrations between the trials, but this was found to be not significant and removed from the model. Temperature was found to have a significant effect on IVCH-based clearance rates ($p = 0.0143$, $n = 8$), and conducting a Tukey HSD post-hoc test found the 10°C and 30°C comparison was significant but no other comparisons were found significant (Fig. 3, **Table 1**).

Table 1. Pairwise comparisons for clearance rates. IVCH p-value results from Tukey HSD test.

Comparison	p - value
22°C-10°C	0.07192
30°C-10°C	0.01421
30°C-22°C	0.72797

Salinity experiment:

Due to uncertainty created by a data collection error in Trial 2, all IVCH data from the T2 (2 hours) time point were removed. The presence of mussels was found to have a significant effect on IVCH levels (Fig S4; Kruskal Wallis Test Trial 1 – $p = 1.46\text{E-}12$, $n = 12$; Trial 2 – $p = 3.01\text{E-}08$, $n = 12$). Time had a significant effect on the IVCH levels for the Mussel treatment for Trial 1 (Friedman’s Rank Sum Test $p = 7.49\text{E-}08$, $n = 12$), No Mussel treatment for Trial 1 (Friedman’s Rank Sum Test $p = 1.26\text{E-}07$, $n = 12$), and the Mussel treatment for Trial 2 (Repeated Measures ANOVA $p = 8.25\text{E-}06$, $n = 12$). Pairwise comparisons show that there is a significant effect of all time comparisons of the Mussel treatments of Trial 1 and Trial 2, and the No Mussel treatment of Trial 1 (**Table S3, Table S4**). For the Trial 1 No Mussel treatment, most of the comparisons were significant except for the T1 and T2 comparisons, and the T3 and T4 comparison (**Table S3**). Salinity did not have a significant effect on the IVCH-based clearance rate for either trial (Fig. 4). Pairwise comparisons also did not show any significant difference between the clearance rates at different salinities (**Table S5**).

Discussion

Urban estuaries have high nutrient inputs and are greatly modified physical environments. This makes it hard to use standard nature-based solutions, like constructed oyster reefs and tidal wetlands, to remove nutrients from the estuary. There is therefore a need for innovative ways to remove nitrogen and mitigate the resulting algae blooms in these often deep and structure-poor urban waterways. Our research shows that *M. leucophaeata* shows promise to contribute to a solution. We established a reproducible method for collecting and acclimating naturally-recruited *M. leucophaeata* into a laboratory setup. This process allowed the animals to re-attach to new substrates that can be oriented similar to how they would be positioned in the environment. The mussels were resilient to the manipulations in the laboratory, just as they appear to be hardy in the stressful environment of Baltimore Harbor.

This investigation showed that *M. leucophaeata* was able to remove phytoplankton at all temperatures tested, as indicated by decreasing IVCH and extracted chlorophyll readings. We observed a significant effect of temperature on the clearance rate, with the clearance rate at 10°C being significantly lower than the clearance rates at 30°C but not 22°C. The clearance rates at 22°C and 30°C were not significantly different from each other, though the data suggests that clearance rates increased slightly between 22°C

and 30°C. In the Netherlands, Rajagopal et al. (2005), found that *M. leucophaeata* particle filtration rate increased until about 27°C and decreased at higher temperatures. However, our chlorophyll clearance rates showed no decline between 22°C and 30°C and appeared to be increasing with temperature. When we tested different salinities typical of Baltimore Harbor, from 5 to 15 ppt, we found that salinity did not affect clearance rates and that mussels showed a significant effect on algae levels at all salinities. In Baltimore Harbor, *M. leucophaeata* occasionally experiences salinities lower than 5ppt (**Fig. S2**), so in the future, it may be important to assess the impact of extreme low salinity on the clearance rates of the mussels.

We observed that *M. Leucophaeata* significantly changed phytoplankton levels at different salinities, from 5 to 15 ppt. However, there was no significant effect of salinity on clearance rates. Unlike the temperature experiment, the salinity trial took place at two different times of the year (July and October) with mussels collected from those respective periods. This difference in collection times may have affected the clearance rates, but the mussel condition was not tested nor evaluated in this study. Additionally, the concentrations of phytoplankton varied with the October trial having a higher starting concentration of phytoplankton than the July trial (**Fig. S4**). The salinity in Baltimore Harbor tends to be between 5 and 15 ppt, although there are occasionally salinities outside the range tested (**Fig. S2**), and those extremes may impact the clearance rates of the mussels.

Our calculated clearance rates for *M. leucophaeata* differed from those previously published. We measured clearance rates from approximately 0.2–0.9 L hr⁻¹ per individual mussel across salinity and temperature experiments. This study used monocultures of phytoplankton in the 4–8 micron range. Rajagopal et al. (2005) using a neutral red dye to calculate the clearance rate of *M. leucophaeata* found the individual rate to be 0.055 L hr⁻¹ per individual. Rodrigues (2023) calculated clearance rates using concentrated seston from a hypereutrophic lagoon and found clearance rates to vary from 0.00123–0.0056 L hr⁻¹ depending on the concentration of the seston. Our clearance rates ranged from approximately 0.275 to 1.06 L hr⁻¹ per individual mussel across the salinity and temperature experiments. Part of this variation in clearance rates could be attributed to the different materials used in the clearance experiment. Rajagopal et al. (2015) used a filtrate of less than 0.45 micron, whereas Rodrigues et al. (2023) used concentrated seston with both organic and inorganic material of 180 micron or smaller. The relatively low Rodrigues clearance rates could be due to the higher initial concentration of filtered material or to the nonspecific nature of the particles. In experiments where bivalves are provided with a high concentration of particles, clearance rates decrease (Wong and Cheung 1999; Lei et al. 1996; Tuttle-Raycraft and Ackerman 2019). Kreeger et al. (2018), in a literature review, compared clearance rates of alternative suspension-feeding bivalves and reported that *M. leucophaeata* has clearance rates comparable to the related mussel, *Dreissena polymorpha*. Further, the extracted chlorophyll values match with what is typically seen in Baltimore Harbor during an algae bloom (50 µg chlorophyll a L⁻¹). *M. leucophaeata* is yet to be tested with a natural community of phytoplankton and inorganic matter.

These results show that *M. leucophaeata* can significantly reduce cultured algae levels. Our study did not examine the ability of *M. leucophaeata* to reduce algae levels of natural blooms and in varying natural settings, which may alter feeding behavior. Natural algae blooms are comprised of different species of varying sizes. In Baltimore Harbor, one of the most common small organisms (~ 6 microns) is the dinoflagellate *Prorocentrum minimum*; one of the larger (~ 50 microns) is *Akashiwo sanguinea* (personal communication, Taylor Armstrong, UMCES). *M. leucophaeata* is a small mussel and while there has not been any literature on this species to date, studies conducted on other bivalve species suggest that there might be some physiological constraints to the size particles that they can ingest (Rosa et al. 2018; Shumway et al. 1985). Other species of bivalves are also known to alter their filter feeding in response to phytoplankton species (Binzer et al. 2018; Galimany et al. 2017). Future work should examine the feeding behavior of natural blooms and varying species and sizes of phytoplankton.

Additional work should focus on the ability of *M. leucophaeata* to reduce algae bloom species in Baltimore Harbor and examine the effects of additional environmental factors. Dissolved oxygen in Baltimore Harbor can be depressed even near the surface due to microbial activity and stratification (Wicks et al. 2011). There is ample literature on the changes in feeding behavior of bivalves experiencing the stress of low oxygen (Kamermans and Saurel 2022; Tang and Riisgård 2018; Widdows et al. 1989). Future work should examine the effect of dissolved oxygen on mussel feeding behavior as it does fall below 2 mg/L in the summer months. Additionally, the experiments we conducted did not use algae blooms collected from the wild. The variety of phytoplankton found within natural algae blooms may affect feeding behavior of the mussels and thus their ability to clear the water.

Bivalve nature-based solutions to algae blooms and provisioning of other ecosystem services are being explored in urban estuaries across the globe. The Bronx River Estuary, located in New York harbor, is closed to shellfish harvest due to high bacteria contamination. Galimany et al. (2017) looked at using the non-commercial ribbed mussel, *Guekensia demissa*, for nutrient removal as an alternative to the eastern oyster. They found that *G. demissa* could remove nutrients from the water, but *G. demissa* does not naturally recruit in that area, which makes implementation of this practice difficult. In the Pacific Northwest, the Budd Inlet in Puget Sound is another area that is also under a harvest advisory for shellfish due to pollution and poor water quality. However, the native bay mussel, *Mytilus trossulus*, does naturally recruit in that area and has been studied for nutrient removal via biomass harvest (Pacific Shellfish Institute 2014). Additionally, the harvested biomass was turned into compost. In Brazil, *M. leucophaeata*, while invasive, has been studied as for its ability to remove seston and other suspended particles from the water (Neves et al. 2020; Rodrigues et al. 2023). Dark False Mussels, which are native to Baltimore Harbor, naturally recruit to substrates and exist with a diverse biofouling community of bryozoans, barnacles, and other particulate feeders (Rome et al. 2023). Preliminary studies show that this community carries out denitrification similar to those measured in intact oyster reef communities (Schott and Cornwell, unpublished; Kellogg et al. 2013). While Baltimore has initiatives to remove nutrients in the waterways, none of the solutions look toward in-water removal strategies as demonstrated in the Bronx River Estuary or Puget Sound. Our results show that there is the potential for

in-water nutrient removal through the filter-feeding of the Dark False Mussel, and there is the option of further nutrient removal through a biomass harvest.

Overall, these results show *M. leucophaeata* can consume laboratory-grown algae levels across salinities and temperatures that are relevant to temperate estuaries. These results show that *M. leucophaeata* can reduce phytoplankton levels and may have potential for the bioextraction of nutrients for a nutrient trading system like the system for oysters in the Chesapeake (Kellogg et al. 2013; Cornwell et al. 2016). To develop such a system, future work should focus on *M. leucophaeata* feeding ability with natural algae blooms and quantification of nitrogen removal through filter feeding.

Declarations

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Figures

Fig. 1

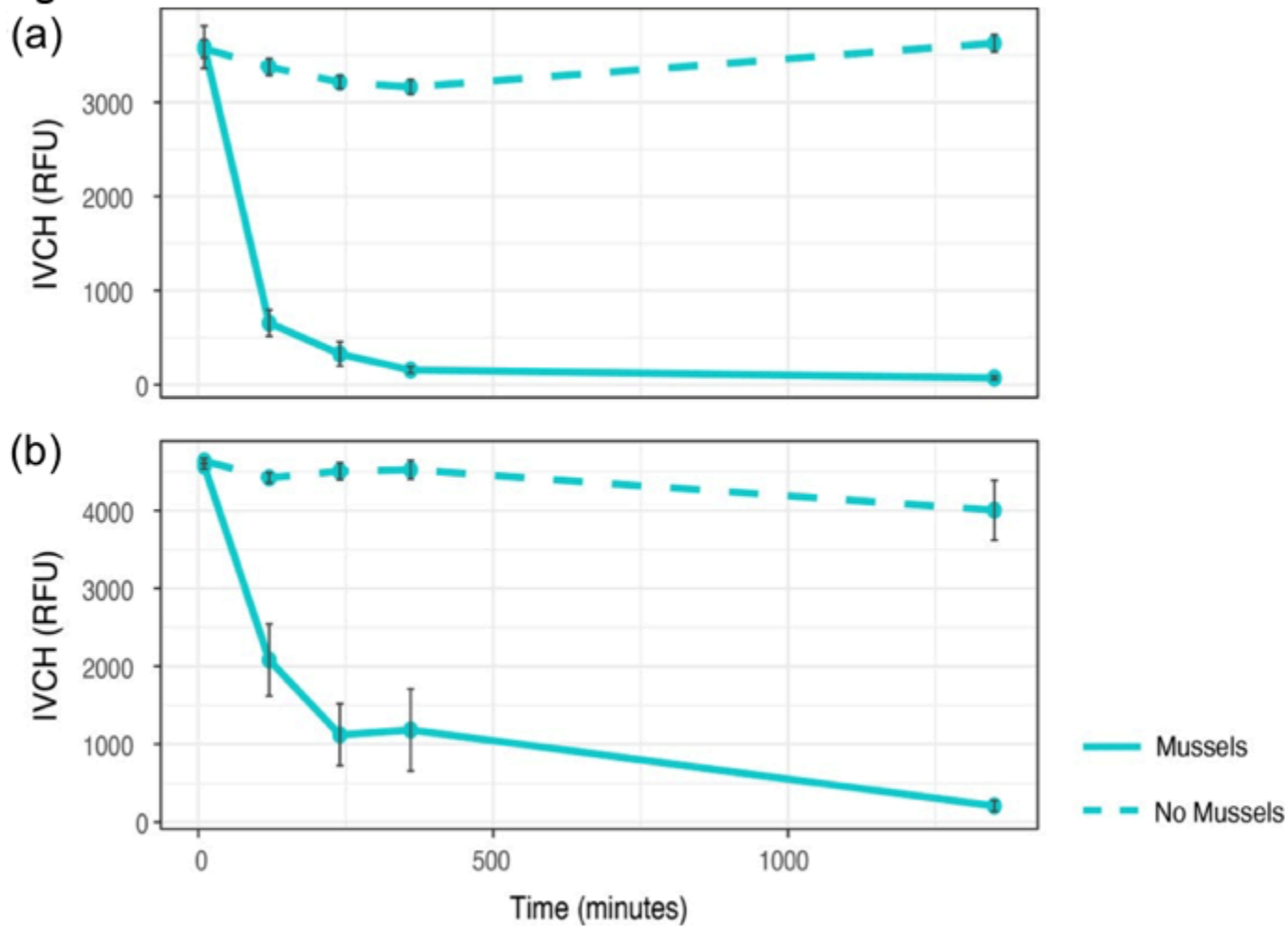


Figure 1

Measured average IVCH over time for **a)** *Isochrysis* and **b)** *Chaetoceros*. Error bars are standard error means. The dashed line represents the No Mussel treatment and the solid line represents the Mussel treatment. Points represented are the T1 (10 minutes), T2 (2 hours), T3 (4 hours), T4 (6 hours), and T5 (22.5 Hours)

Fig. 2

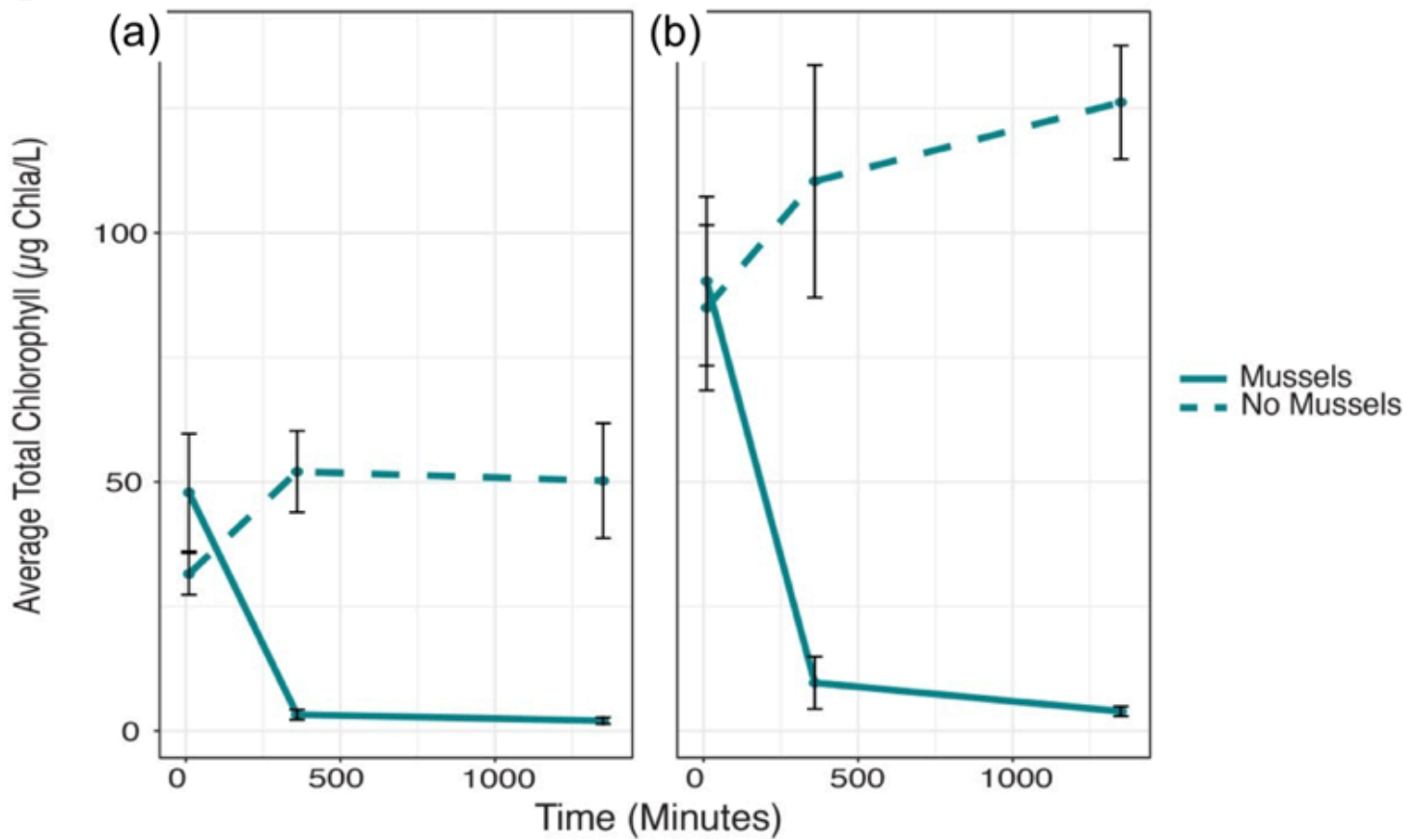


Figure 2

Average total extracted chlorophyll over time of cultures with **a)** *Isochrysis* and **b)** *Chaetoceros*. Error bars are standard error means. Solid lines represent the Mussel treatment (*Isochrysis* n = 3 per timepoint, *Chaetoceros* n = 4 per timepoint) and the dashed lines represent the No Mussel treatment (*Isochrysis* n = 4 per timepoint, *Chaetoceros* n = 4 per timepoint). Points are T1 (10 minutes), T4 (6 hours), and T5 (22.5 hours)

Fig. 3

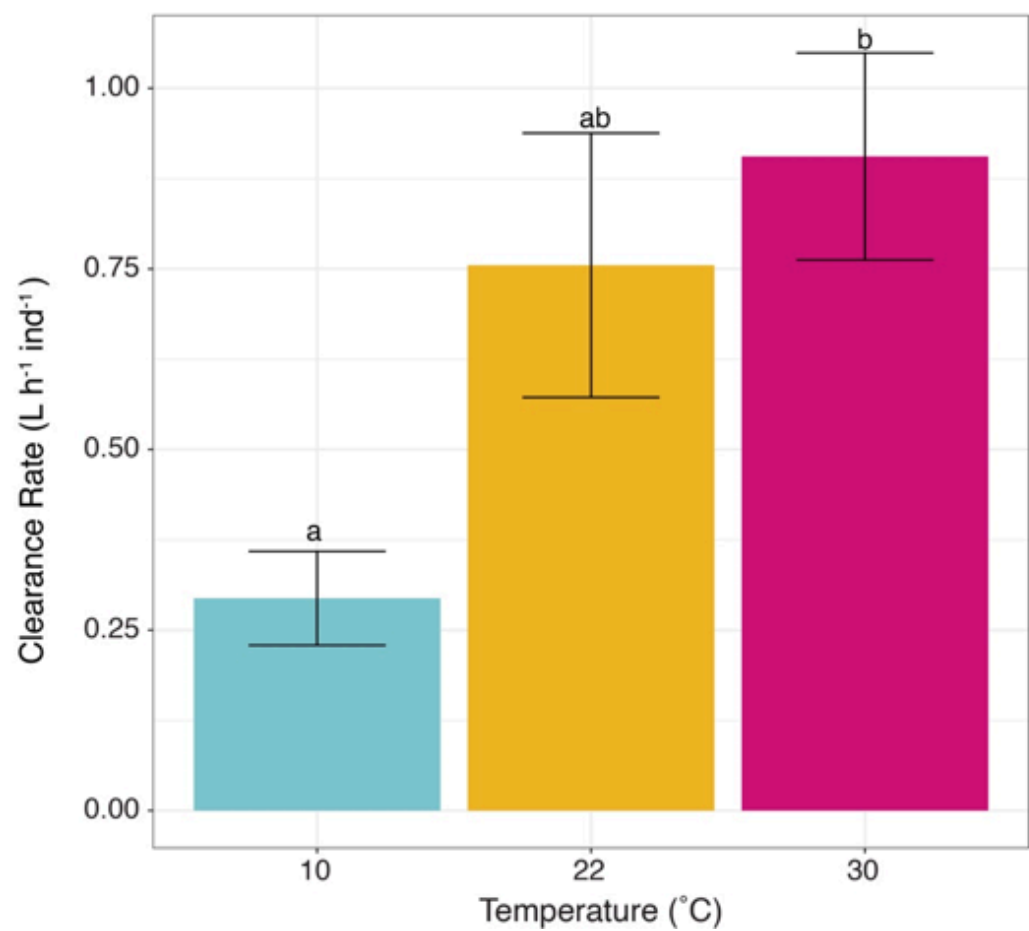


Figure 3

Clearance rates per mussel for the three temperatures tested. The letters above columns indicate the significance of pairwise comparison from the Tukey HSD test from Table 1. Error bars are standard error means

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